Road Accident Analysis Report (2021–2022)

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Tools Used (Python [Pandas, Numpy, Matplotlib, Seaborn)

PROJECT OVERVIEW

Objective: Identify **patterns, risk factors, and prevention strategies**

- Focus area: Time trends
- Geographic hotspots
- Environmental conditions
- Driver behaviour
- **307,973 accidents** and **417,883 casualties**

EXECUTIVE SUMMARY

- 307K accidents | 563K vehicles | 417K casualties
- ~85% are slight severity
- Majority occur in urban areas
- Peak: evenings, Fridays, Oct–Nov

Implication:
Accidents are driven by **high exposure, not extreme conditions**

GEOGRAPHIC INSIGHT

Top High-Risk Areas

- Birmingham
- Leeds
- Manchester
- Bradford
- Westminster

Implication:
Urban areas = **higher traffic density → higher accident risk**

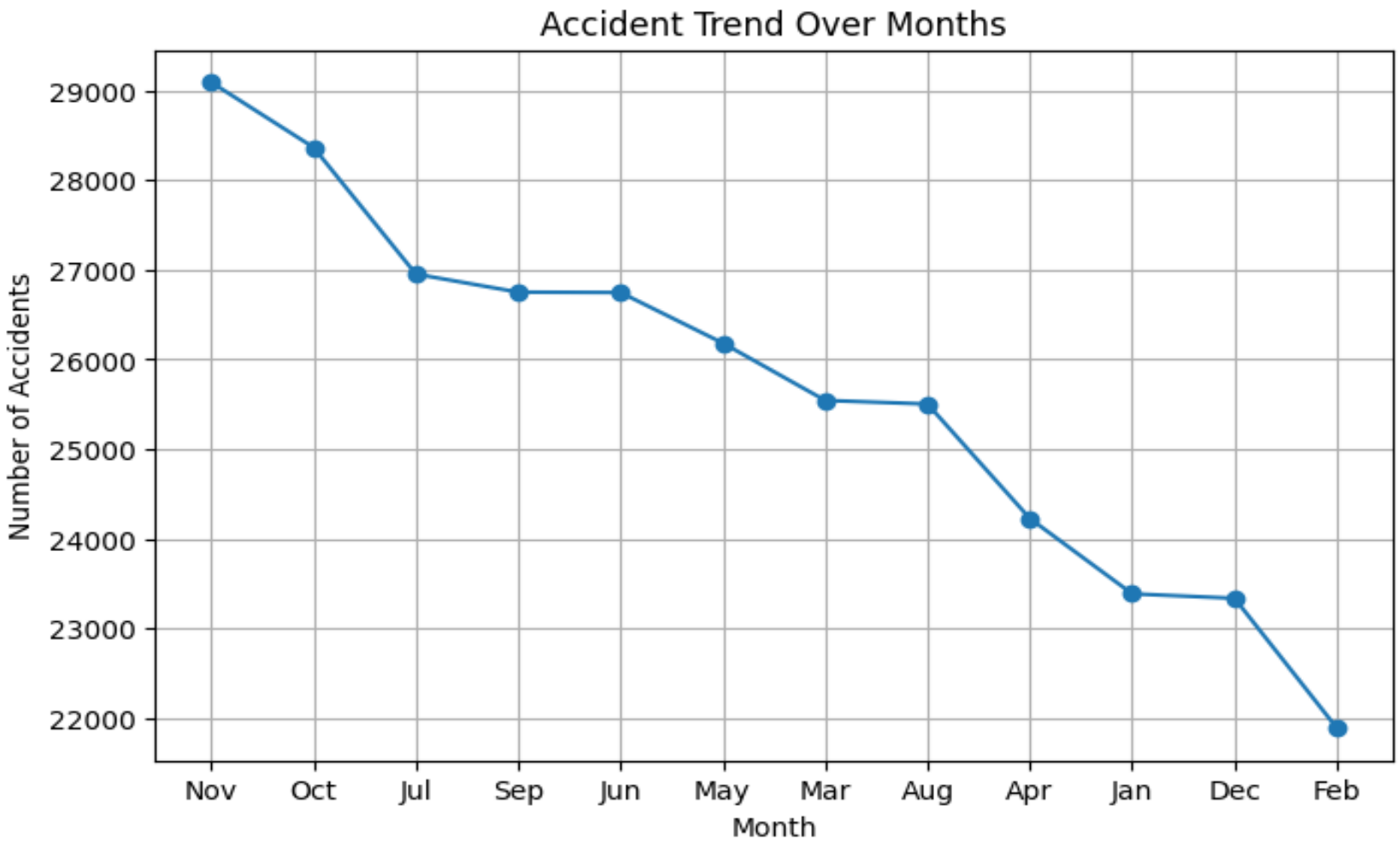
TEMPORAL TRENDS

- 2021: 163,554
- 2022: 144,419

Peak Months:

- November
- October
- July

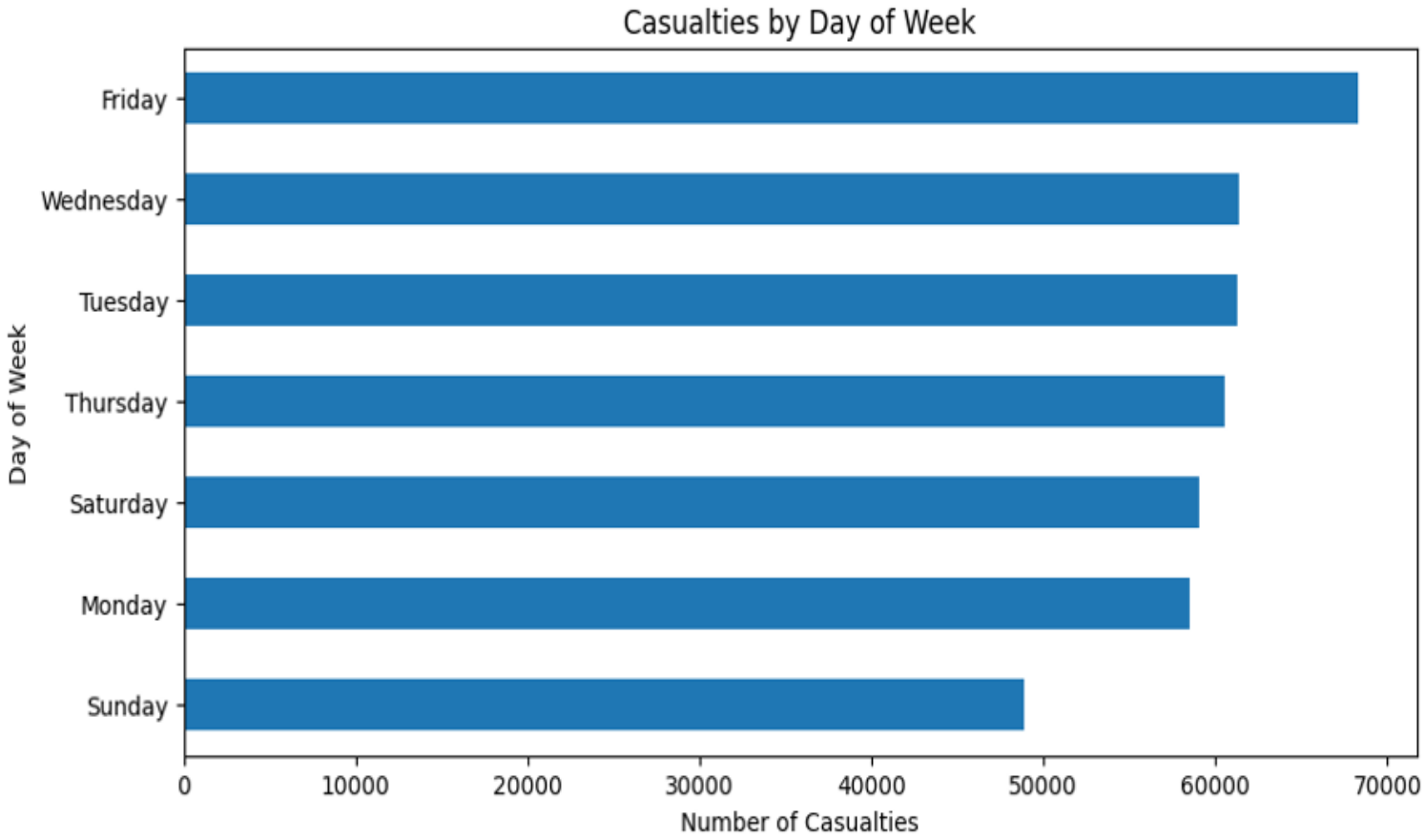
Implication:
Late-year spikes due to weather + reduced daylight + travel


Accident Trend Over Months
Number of Accidents
29000
28000
27000
26000
25000
24000
23000
22000
Nov
Oct
Jul
Sep
Jun
May
Mar
Aug
Apr
Jan
Dec
Feb
Month

WEEKLY PATTERNS

- Friday = highest accidents
- Midweek (Tue–Wed) also high
- Sunday = lowest

Implication:
Fatigue + increased movement drive end-of-week risk


Casualties by Day of Week
Friday
Wednesday
Tuesday
Thursday
Saturday
Monday
Sunday
Day of Week
0
10000
20000
30000
40000
50000
60000
70000
Number of Casualties

TIME OF DAY ANALYSIS

Peak Hours:

- 3 PM – 6 PM
- Highest at 17:00

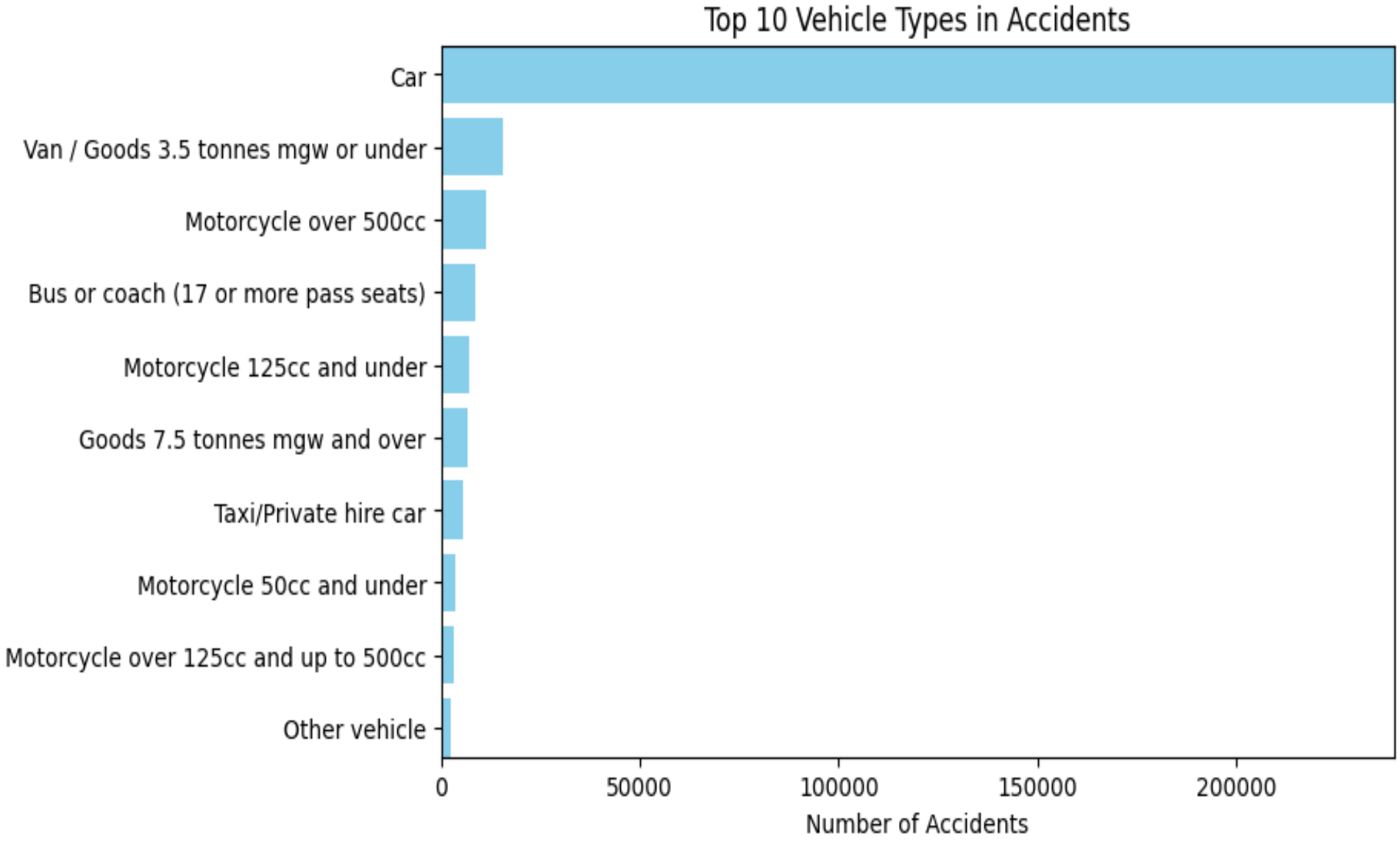
Implication:
Rush hour = highest congestion + accident probability

VEHICLE INVOLVEMENT

- Cars dominate (325K+)
- Vans & motorcycles follow

Implication:

- Cars → volume-driven risk
- Motorcycles → higher relative vulnerability


Top 10 Vehicle Types in Accidents
Car
Van / Goods 3.5 tonnes mgw or under
Motorcycle over 500cc
Bus or coach (17 or more pass seats)
Motorcycle 125cc and under
Goods 7.5 tonnes mgw and over
Taxi/Private hire car
Motorcycle 50cc and under
Motorcycle over 125cc and up to 500cc
Other vehicle
0
50000
100000
150000
200000
Number of Accidents

ENVIRONMENTAL CONDITIONS

Weather:

- Most accidents in fine weather

Road:

- Dry roads dominate
- Wet roads → higher severity

Implication:
Risk is influenced by behaviour, not just conditions

URBAN VRS RURAL

- Urban: 255K
- Rural: 162K

Implication:
Urban areas = intersections + congestion = higher risk

SPEED ANALYSIS

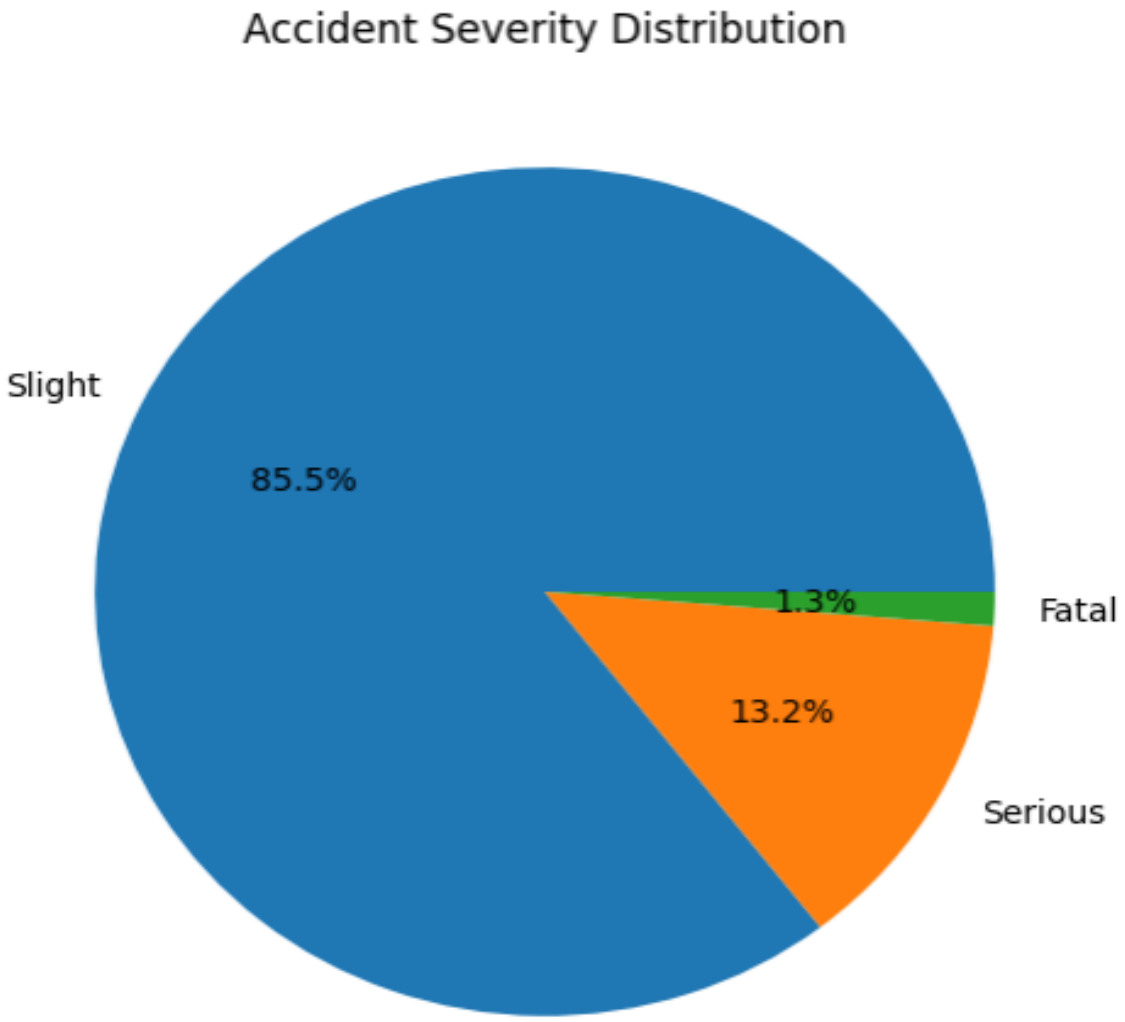
- 30 mph zones = highest accidents
- Higher speeds = fewer but more severe

Implication:
Urban low-speed zones are high-frequency risk areas

ACCIDENT SEVERITY

- Slight: ~85%
- Serious: 59K
- Fatal: 7K

Implication:
Most are minor, but fatalities remain preventable


Accident Severity Distribution
Slight
85.5%
1.3%
Fatal
13.2%
Serious

KEY RISK PATTERNS

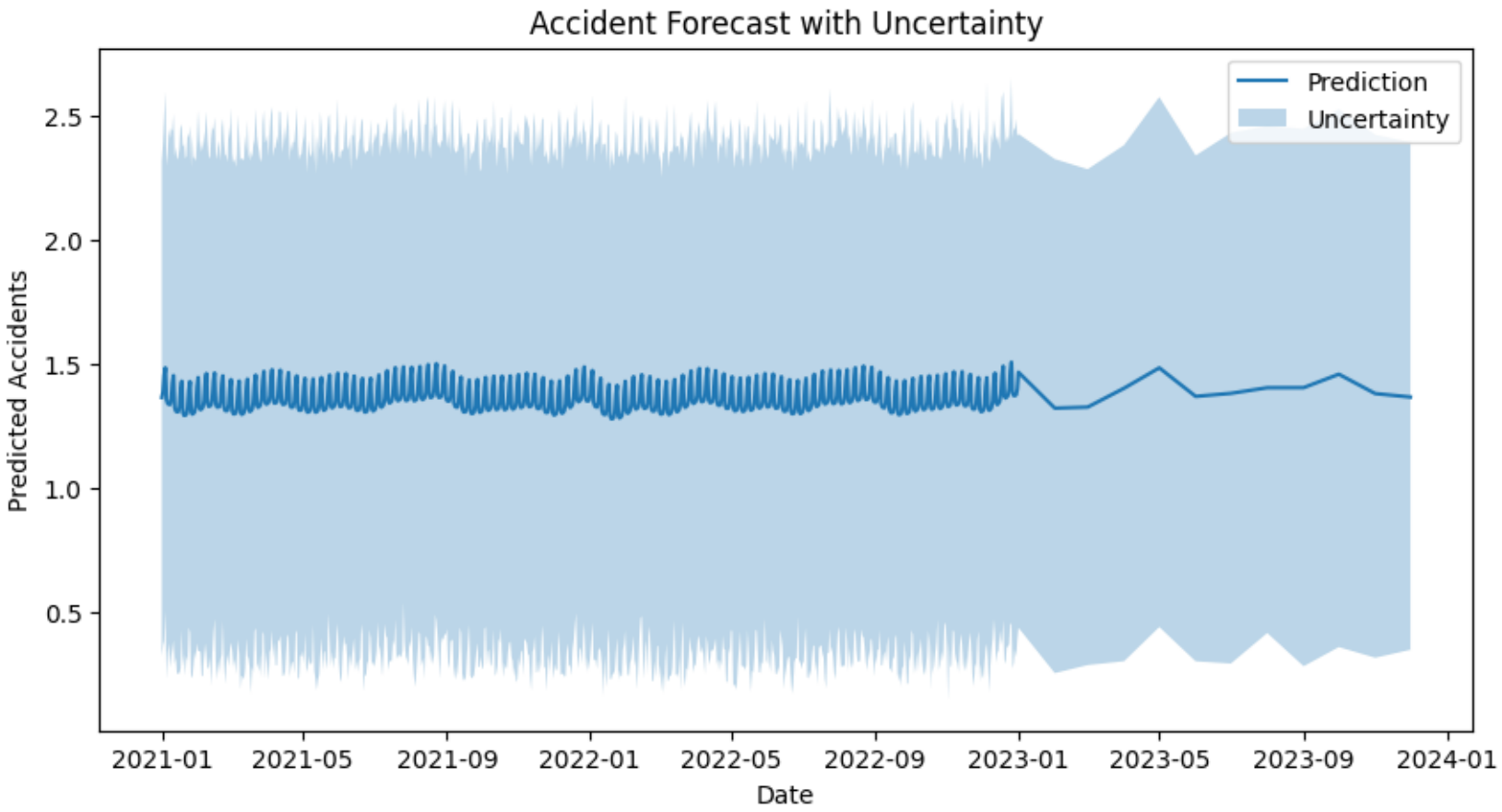
High-Risk Scenario:

- Urban + 30 mph + Car + Daylight + Evening

Key Drivers:

- Congestion
- Rush hour
- Seasonal trends
- Human factors (fatigue, distraction)

FORECAST


Accident Forecast with Uncertainty
Prediction
Uncertainty
Predicted Accidents
2.5
2.0
1.5
1.0
0.5
2021-01
2021-05
2021-09
2022-01
2022-05
2022-09
2023-01
2023-05
2023-09
2024-01
Date

RECOMMENDATIONS

Urban Traffic & Infrastructure

- Optimize traffic light timing during rush hours (3–6 PM)
- Introduce dedicated turning lanes at high-risk intersections
- Deploy real-time traffic monitoring systems in major cities
- Improve road markings and signage visibility, especially in busy zones

Driver Behaviour & Enforcement

- Increase police presence on Fridays and evening hours
- Enforce strict speed monitoring in 30 mph zones (speed cameras)
- Launch targeted campaigns on:
  - Distracted driving (mobile phone use)
  - Fatigue during end-of-week driving
- Introduce penalties for repeat traffic offenders

RECOMMENDATIONS II

Rush Hour Risk Reduction

- Encourage **flexible work hours / staggered commuting**
- Promote **public transport and carpooling** to reduce congestion
- Use **live traffic alerts** via mobile apps to redirect drivers

Weather & Seasonal Preparedness

- Issue **real-time weather-based driving alerts**
- Pre-treat roads and improve drainage in **rain-prone areas**
- Increase enforcement and awareness during:
  - **October–November peaks**
  - Holiday travel periods

RECOMMENDATIONS III

High-Risk Road Users (Motorcycles)

- Mandatory **protective gear enforcement (helmets, reflective wear)**
- Create **dedicated motorcycle lanes** where possible
- Offer **subsidized rider safety training programs**

Data & Smart City Solutions

- Implement **AI-powered accident prediction systems**
- Use historical data to identify and monitor **accident hotspots**
- Install **smart cameras** for real-time violation detection
- Continuously update policies based on **data insights**

RECOMMENDATIONS IV

Public Awareness & Education

- Introduce **road safety education in schools and workplaces**
- Run **localized awareness campaigns** in high-risk districts
- Use social media for **targeted safety messaging during peak periods**

Emergency Response Improvement

- Improve **ambulance response time in urban hotspots**
- Position emergency units closer to **high-accident zones**
- Use GPS systems to **optimize emergency routing**

Thank you!!!